LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Third laboratory (Variant 3.1)

Interrupts manipulators of the 32 bit microcontrollers

Mandatory part (6 points):

- Make the red led to shine when the user button is pressed. The red led should remain to shine when the user button is released. Use the external interrupt EXTI0 alternative function in order to capture then the button is pressed. Capture the falling edge of the external signal (**2 points**).
- Write the program in order to change the blinking frequency of the green led. The initial blinking frequency should be equal to 0.5 second. The blinking frequency should vary periodically between 0.5, 1.0 and 1.5 seconds. Implement blinking using the *delay* function in an infinite while loop. The blinking frequency should be controlled with the user button. Capture a button press moment using external interrupts functionality. Capture the falling edge of the external signal (**4 points**).

Additional parts (4 points):

- Make the configuration of the timer TIM2 and write the program in order to toggle the green led every 1 second (**2 points**).
- Make the configuration of the timer TIM3 and write the program in order to toggle the red led every 2.5 second (**2 points**).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Third laboratory (Variant 3.2)
Interrupts manipulators of the 32 bit microcontrollers

Mandatory part (6 points):

- Make the green led to shine when the user button is pressed. The green led should remain to shine when the user button is released. Use the external interrupt EXTI0 alternative function in order to capture then the button is pressed. Capture the falling edge of the external signal (**2 points**).
- Write the program in order to change the blinking frequency of the red LED. The initial blinking frequency should be equal to 0.5 second. The blinking frequency should vary periodically between 0.5, 1.5 and 3.5 seconds. Implement blinking using the *delay* function in an infinite while loop. The blinking frequency should be controlled with the user button. Capture a button press moment using external interrupts functionality. Capture the falling edge of the external signal (**4 points**).

Additional parts (4 points):

- Make the configuration of the timer TIM2 and write the program in order to toggle the red led every 1 second (**2 points**).
- Make the configuration of the timer TIM3 and write the program in order to toggle the green led every 3.5 second (**2 points**).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Third laboratory (Variant 3.3)
Interrupts manipulators of the 32 bit microcontrollers

Mandatory part (6 points):

- Make the green led to shine when the user button is pressed. The green led should remain to shine when the user button is released. Use the external interrupt EXTI0 alternative function in order to capture then the button is pressed. Capture the falling edge of the signal (**2 points**).
- Write the program in order to change the blinking frequency of the red led. The initial blinking frequency should be equal to 1 second. Implement the blinking using the *delay* function in an infinite while loop. The blinking frequency should be controlled with the user button. The blinking frequency should increase by 1.5 seconds each time the user button is pressed. The blinking frequency should return to 1 second when the maximum blinking frequency of 4 seconds will be reached. Capture a button press moment using external interrupts functionality. Capture the falling edge of the external signal (**4 points**).

Additional parts (4 points):

- Make the configuration of the timer TIM2 and write the program in order to toggle the red led every 0.5 second (**2 points**).
- Make the configuration of the timer TIM3 and write the program in order to toggle the green led every 2 second (**2 points**).

LABORATORY OF DESIGN OF COMPUTERIZED SYSTEMS

Third laboratory (Variant 3.4)
Interrupts manipulators of the 32 bit microcontrollers

Mandatory part (6 points):

- Make the red led to shine when the user button is pressed. The red led should remain to shine when the user button is released. Use the external interrupt EXTI0 alternative function in order to capture then the button is pressed. Capture the falling edge of the signal (**2 points**).
- Write the program in order to change the blinking frequency of the green LED. The initial blinking frequency should be equal to 2.5 seconds. Implement the blinking using the *delay* function in an infinite while loop. The blinking frequency should be controlled with the user button. The blinking frequency should decrease by 0.5 second each time the user button is pressed. The blinking frequency should return to initial 2.5 seconds when the minimum blinking frequency of 0.5 second will be reached. Capture a button press moment using external interrupts functionality. Capture the falling edge of the external signal (**4 points**).

Additional parts (4 points):

- Make the configuration of the timer TIM2 and write the program in order to toggle the red led every 1 second (**2 points**).
- Make the configuration of the timer TIM3 and write the program in order to toggle the green led every 0.5 second (**2 points**).